



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SCSE2243 APPLICATION DEVELOPMENT PROJECT

SoFea Management System

Phase 0 – Problem Framing Document

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1. Domain Selection

The project is developed under the management domain, focusing on building an integrated system known as the SoFea Management System. The system aims to support multiple administrative functions such as activity management, financial management, and operational coordination within a single platform.

2. Problem Description

Currently, many organizations rely on multiple separate tools and manual processes to manage their administrative operations. For example, activity planning may be handled using spreadsheets, financial records are stored in different systems, and communication is often done through messaging platforms. This leads to fragmented systems, where data is scattered and not centralized.

In addition, many processes are still manual and inefficient, requiring repeated data entry, increasing the risk of human error, and making it difficult to track progress or generate accurate reports. The lack of integration between systems also results in poor data consistency and limited visibility, which affects decision-making.

Therefore, there is a strong need for a centralized management system that can integrate these functions into a single platform. Such a system would improve efficiency, reduce redundancy, and provide better data organization and accessibility for users.

3. Existing System Comparison

Existing systems used for organizational management can be broadly categorized into task management tools, financial systems, and enterprise resource planning (ERP) systems. Task management platforms such as Trello and Asana provide effective features for task assignment, progress tracking, and team collaboration. However, they lack financial management capabilities and do not offer full system integration. On the other hand, financial systems like QuickBooks are designed specifically for financial tracking and reporting, but they do not support activity or task coordination. More comprehensive solutions such as Odoo and SAP ERP integrate multiple business functions into a single platform, including finance, operations, and project management. Despite their powerful capabilities, these systems are often complex, costly, and require significant time to learn, making them less suitable for small organizations or student-level projects.

Therefore, the proposed SoFea Management System aims to bridge this gap by providing a centralized, user-friendly, and integrated platform that combines activity management, financial tracking, and task coordination in a simpler and more accessible way.

4. System Idea

The SoFea Management System is an integrated digital platform designed to support and streamline various administrative and operational functions within an organization. It combines multiple features such as activity management, financial tracking, and task coordination into a single unified system.

The system is used to help organizations manage their daily operations more efficiently by centralizing data, reducing reliance on multiple disconnected tools, and improving communication among users. It allows users to plan activities, monitor financial transactions, assign tasks, and access important information in one place, making management processes more organized and effective.

5. Objectives

The main objectives of the SoFea Management System are:

- 1.To improve operational efficiency by reducing manual work and automating repetitive tasks
- 2.To centralize management processes into a single platform for better organization
- 3.To minimize human errors through structured data handling and system support
- 4.To enhance decision-making by providing accurate and real-time information
- 5.To improve data integration and accessibility across different functions

6. 6.1 Entity Relationship Diagram (ERD)

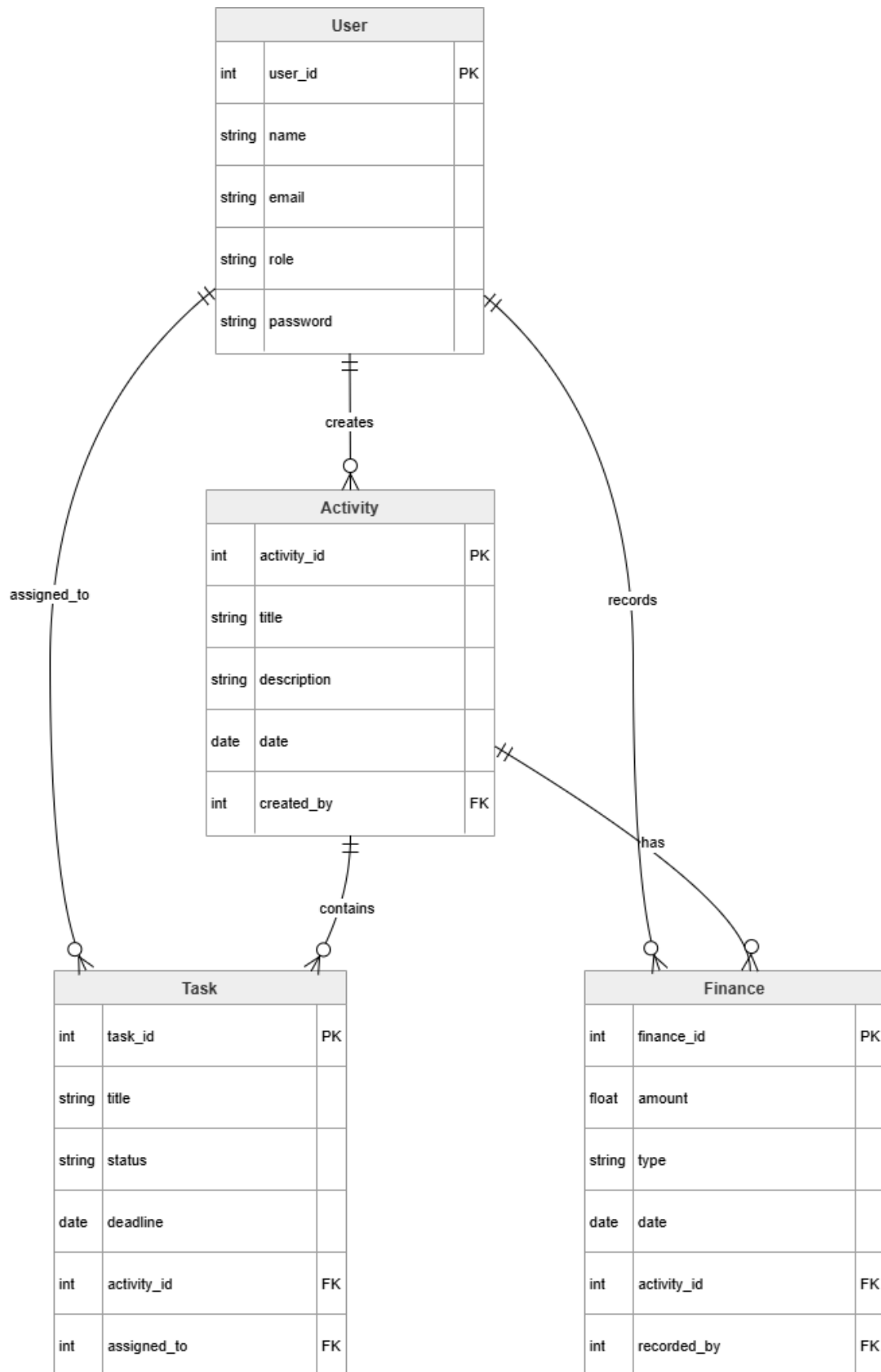
The Entity Relationship Diagram (ERD) illustrates the data structure of the SoFea Management System. It defines the main entities, their attributes, and the relationships between them.

The system consists of four core entities: User, Activity, Task, and Finance. Each entity represents a key component of the system's functionality.

A User can create multiple Activities, establishing a one-to-many relationship. Each Activity can contain multiple Tasks and Finance records, allowing the system to manage both operational and financial aspects in a centralized manner.

The Task entity is associated with both Activity and User, where each task belongs to a specific activity and is assigned to a user. Similarly, the Finance entity is linked to an activity and recorded by a user, ensuring accountability and traceability of financial transactions.

This ERD design supports efficient data organization, reduces redundancy, and ensures strong relationships between different system components.



[Figure 1: ERD of SoFea Management System]